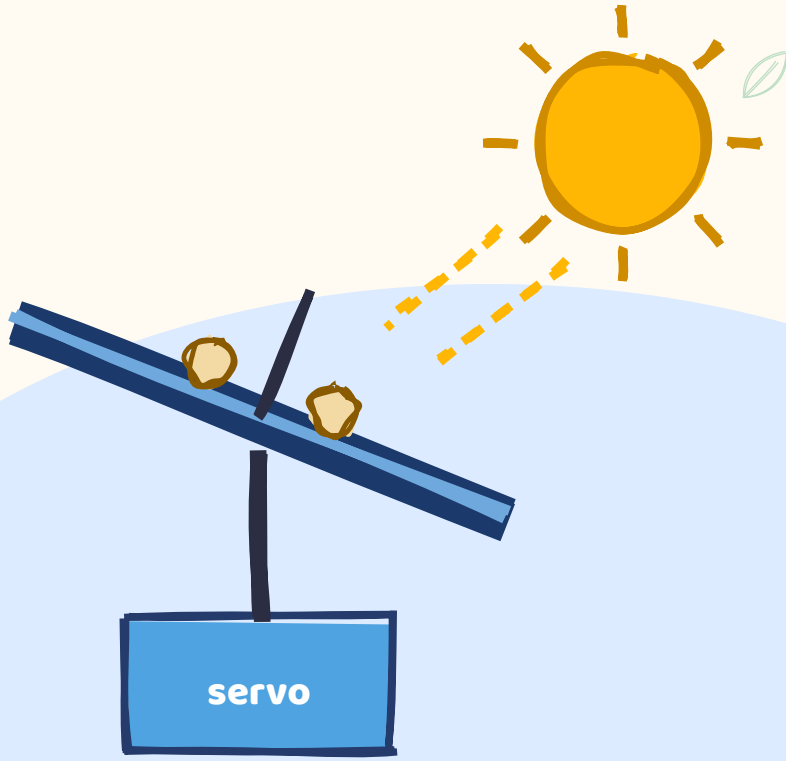
Junior Makers · Class 8

Sun-Tracking Solar Panel

A panel that chases the sun all day long!

Two light sensors **Servo motion** **Clean energy**



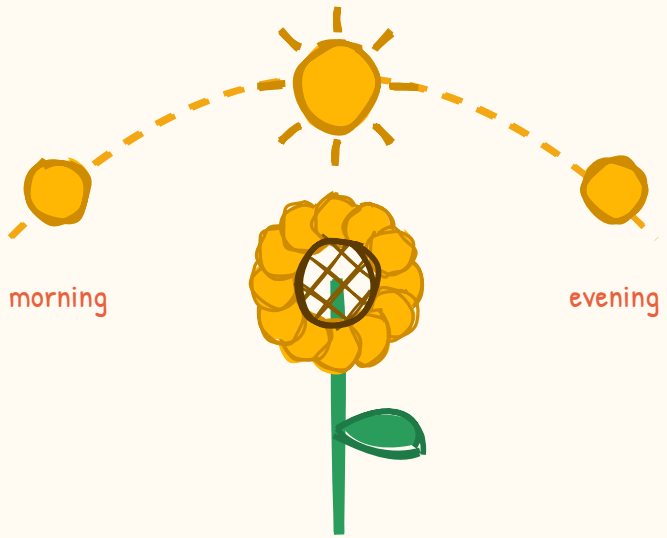
Did you know?
Young sunflowers turn to face the sun as it moves across the sky. Our robot will copy them!

Let's think...

The sun moves. Why doesn't the solar panel?

A puzzle

Most rooftop solar panels are **stuck** in one position. In the morning and evening the sun shines on them **from the side**.
A sunflower would never put up with that!



Move it by hand?

Use lots of panels?

Build a robot that follows the sun!

Today: we build a panel with two light "eyes" that turns itself to face the brightest light.

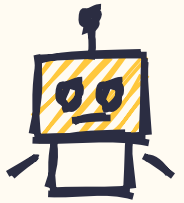
LEARN

Learning aims

Today I will learn to...

1. Explain why a panel makes the **most energy** when it faces the sun
2. Read **two LDR light sensors** and compare them
3. Turn a **servo** towards the **brighter** side
4. Understand **feedback**: sense, compare, move, repeat
5. **Measure** the panel's output with and without tracking

Keywords
solar panel · LDR · servo · compare · feedback · dead zone · current



Watch and think

Video time!



Solar Energy: How Do Solar Panels Work?

learning junction · about 4 minutes · youtu.be/viQInAU0ekc

Watch for...

1. What does a solar panel turn sunlight into?
2. When does a panel make the **most** electricity?
3. Name one place that uses solar power.

Scan to watch

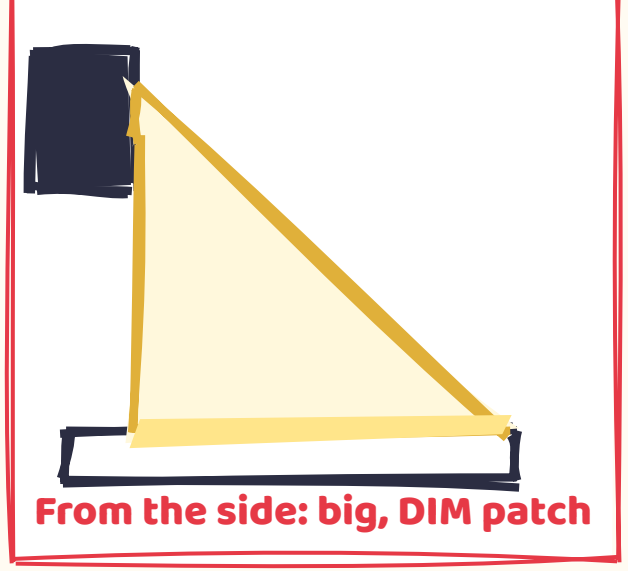
After the video: Why would a panel that **follows** the sun make more electricity than one that stays still?

The science

Straight-on light is strongest



Straight on: small, BRIGHT spot



From the side: big, DIM patch

Try it now!

Shine a torch straight down on paper. Now tilt it. The **same light** gets spread over a **bigger area**, so each part gets less.

Same for solar panels

Sunlight hitting the panel **straight on** gives the most electricity. Sunlight from the side gives much less.

The sun's journey

It rises in the EAST, is highest at midday, and sets in the WEST.

Real world

Who follows the sun?



Solar farms
Rows of panels on trackers make roughly a quarter more energy than fixed ones



Satellites
Turn their solar wings to keep facing the sun in space



Sunflowers
Young plants follow the sun from east to west each day



Pakistan
Has lots of sunshine. Solar panels on rooftops have grown very fast

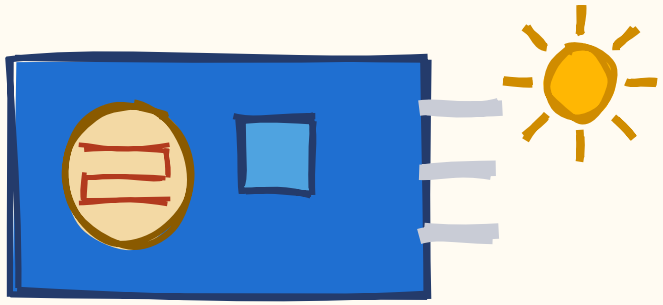


Have you seen solar panels near your home or school? Which way do they **face**? Do they move?

Think about it
A tracker costs more and has moving parts. When is it worth it?

Meet the sensor

The LDR: a light-sensing eye



LDR module

How it works

LDR means **Light Dependent Resistor**. The squiggly track changes how easily electricity flows when light falls on it. The Arduino turns that into a **number**.

We need TWO

One eye on the left, one eye on the right. Then we can compare!

Pin	What it does
VCC	5V
GND	Ground
AO	A number: 0 to 1023

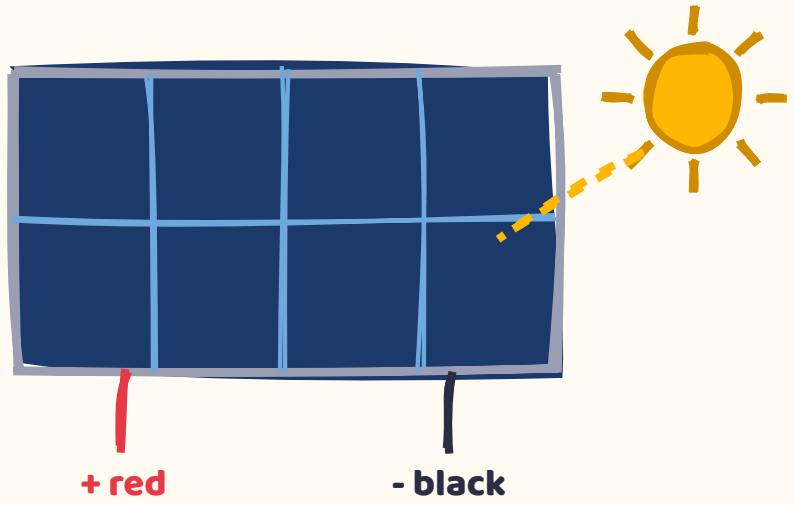
Check YOUR module!

On some modules bright light gives a **bigger** number, on others a **smaller** one. Test with a torch on the Serial monitor.

LEARN

Meet the power maker

The small solar panel



Two different jobs

LDRs = the eyes. They **sense** where the light is.

Solar panel = the power maker. It **makes** electricity.

Servo = the neck. It **turns** the panel.

A panel is made of **solar cells** (the squares). Light knocks tiny particles called **electrons** loose, and they flow as **electricity**. No fuel, no smoke, no noise!

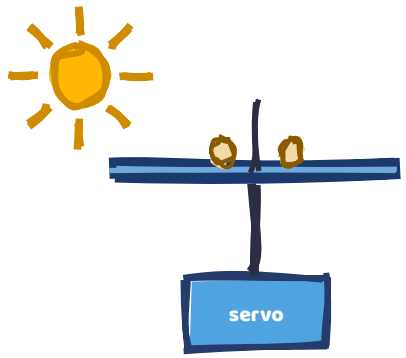
Careful

Glass panels can crack. Hold by the edges and never stand on one.

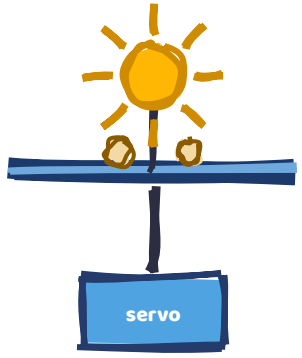
The big idea

Two eyes and a wall between them

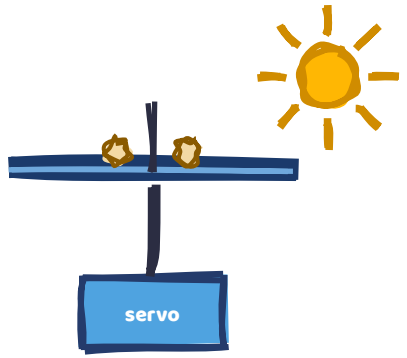
A small **wall** sits between the two LDRs. If the sun is on one side, the wall puts the **other eye in shadow**.



Left is brighter
Turn LEFT



Both the same
Stay still: we are facing the sun!



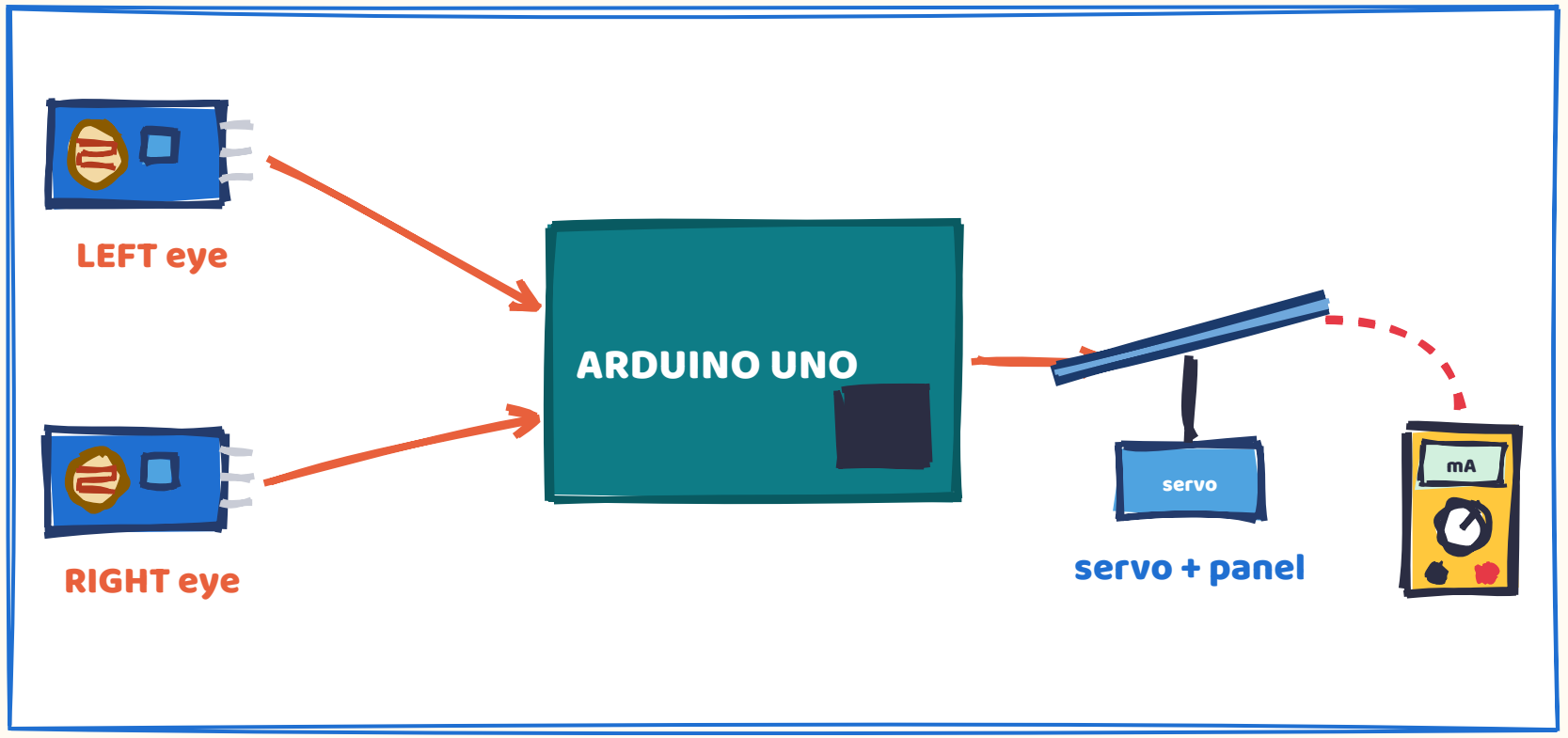
Right is brighter
Turn RIGHT

Feedback loop: sense, compare, move a little, sense again, forever.

Engineer word
Machines that correct themselves use FEEDBACK.

The plan

Sense, compare, move



Two circuits

The **Arduino circuit** moves the panel.
The **panel circuit** goes only to the multimeter, so we can measure the energy it makes.

Why not power the Arduino from the panel?
Our small panel is too weak. Real trackers use a bit of their own power.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Be the sun tracker!
2. Our sun tracker kit (per group)
3. Wiring the sun tracker
4. Make the tracker mount
5. The sun tracker in ArduinoBlocks
6. Don't build a jittery robot!
7. Does tracking make more energy?

Class game

Be the sun tracker!

How to play

1. Two children are the **left eye** and **right eye**. One child is the **servo** holding a book (the panel).
2. The teacher walks around with a torch (the sun).
3. Eyes shout **"Bright!"** or **"Dark!"**.
4. The servo turns a little towards "Bright" each time.
5. When both eyes say "Bright", **freeze!**

Then discuss

What happened when the servo turned **too fast**?

What if both eyes say "Dark" (a cloud or night)?



Big idea

Small steps are better than big jumps. Big jumps overshoot!

Build it

Our sun tracker kit (per group)



Arduino Uno
The brain



2 × LDR module
The two eyes



Servo SG90
The neck



Small solar panel
The power maker



Digital multimeter
Measures the energy



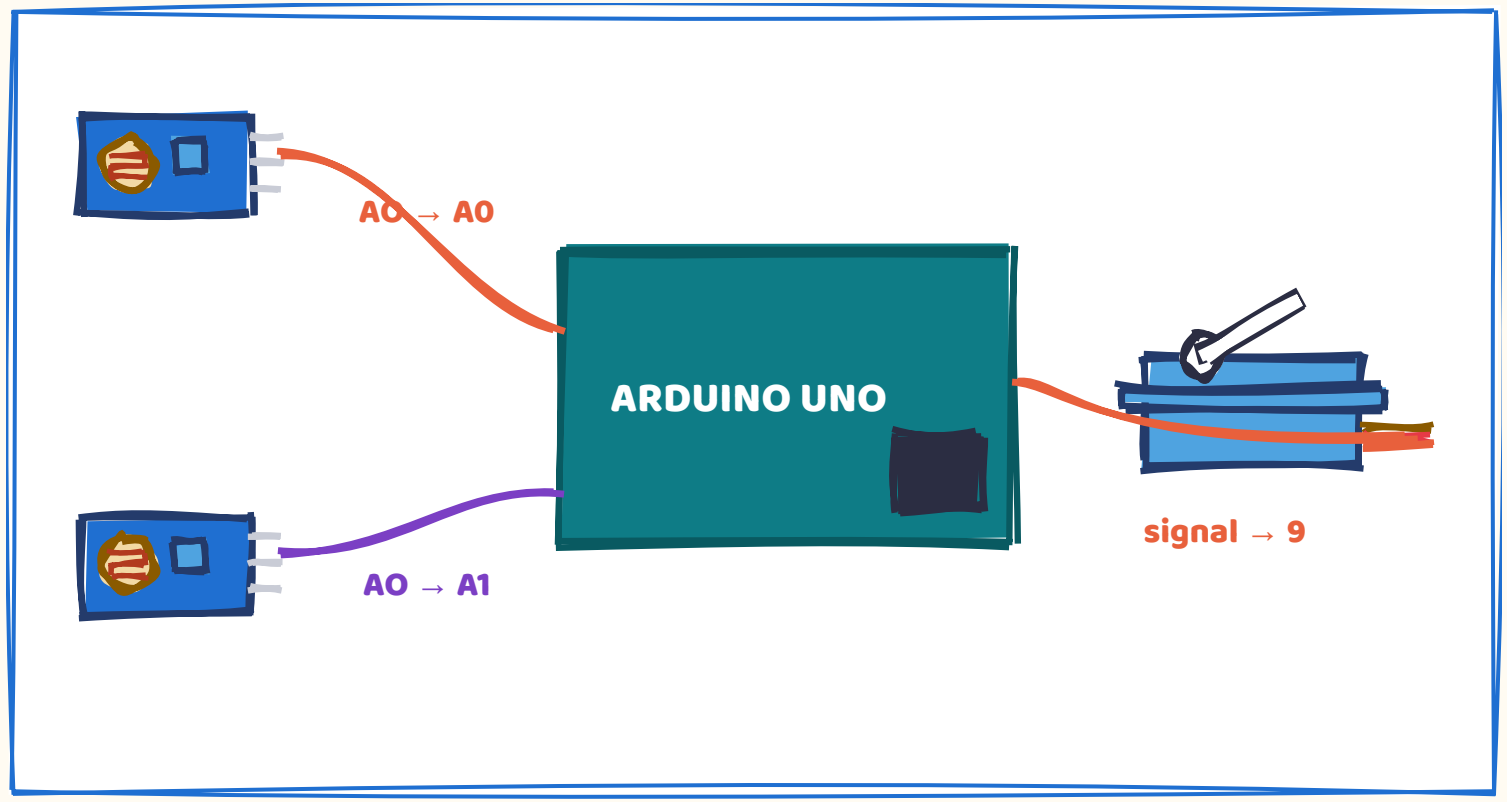
Cardboard + glue
Mount and shadow wall

Also: breadboard, jumper wires, USB cable, laptop, a torch or phone light (our indoor "sun"), tape.

Re-use
The servo skills from the rain-sensing window come back today!

Build it

Wiring the sun tracker



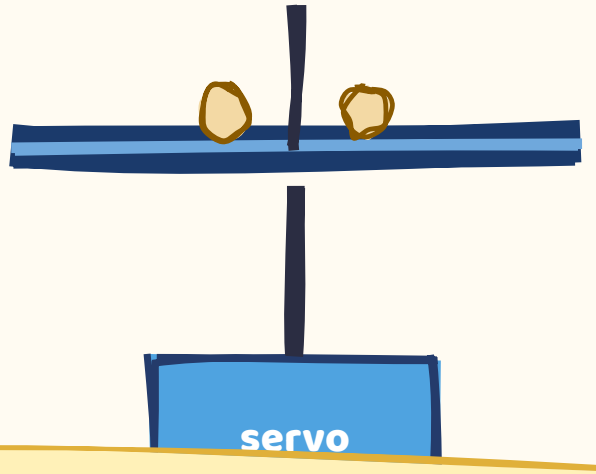
Part	Arduino
Both LDR VCC / GND	5V / GND
LEFT LDR AO	A0
RIGHT LDR AO	A1
Servo brown / red	GND / 5V
Servo orange (signal)	pin 9
Solar panel + / -	multimeter only

Breadboard tip
Use the + and - rails for 5V and GND. Three parts all need them!

Build it

Make the tracker mount

1. **Cut** a card rectangle the size of your panel. Tape the panel on top.
2. **Wall** glue a small card wall (about 4 cm tall) across the middle.
3. **Eyes** tape one LDR on each side of the wall, facing up.
4. **Neck** hot-glue the servo horn under the centre (teacher helps).
5. **Centre** set the servo to 90° BEFORE fixing, so it can turn both ways.



Hot glue
Teacher only. The glue gun tip is hotter than boiling water!

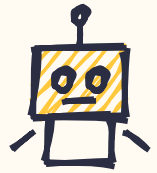
Code it

The sun tracker in ArduinoBlocks

```
when Arduino starts
    set angle to 90 · servo pin 9 → angle
forever
    set left to read analog A0
    set right to read analog A1
    set diff to left - right
    if diff > 50 and angle < 170
        set angle to angle + 2
    else if diff < -50 and angle > 10
        set angle to angle - 2
    servo pin 9 → angle
    wait 30 ms
```

Read it like a story

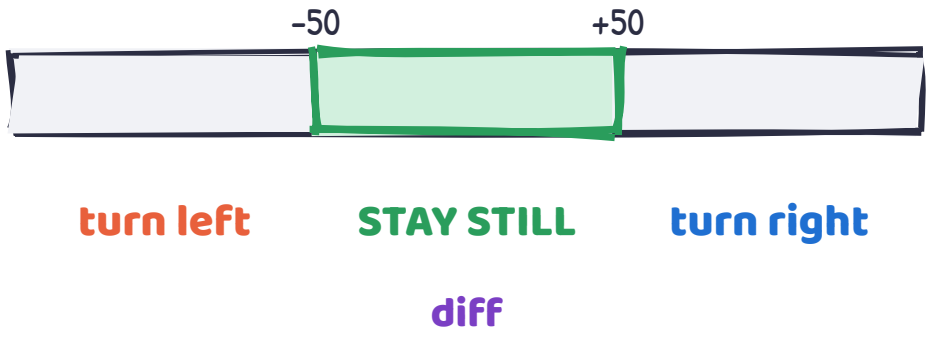
Start in the **middle** (90°).
Find the **difference** between the eyes.
Big difference? Move **2°** that way.
Small difference? **Stay still.**



Runs away from the torch?
Your module counts the other way. Swap A0 and A1 in the code and try again!

Why 50?

Don't build a jittery robot!



Experiment

Change 50 to **5**. What happens?

Change it to **300**. What happens?

Find the **best** number for your tracker.

The dead zone

Two LDRs are never **exactly** the same. Without a dead zone the servo would twitch left, right, left, right forever. The dead zone says: "close enough, stay still".

Real engineers

Air conditioners and car cruise control use dead zones too!

Test like a scientist

Does tracking make more energy?

Test	Mode	Panel current
Panel flat, torch from the LEFT	fixed	_____ mA
Panel flat, torch from the RIGHT	fixed	_____ mA
Tracker ON, torch from the LEFT	tracking	_____ mA
Tracker ON, torch from the RIGHT	tracking	_____ mA

Fair test!

Same torch · same distance (about 30 cm) · same height. Change **only one thing**: tracking on or off.

Meter on mA (current). Red probe to panel +, black to -. Current shows the difference much more clearly than volts.

Teacher note

Small panels give under 200 mA, safe on the meter's 200 mA range. Check the red lead is in the mA socket.

Level up!

Sun tracker challenges

Choose your challenge

- **Night mode:** if both eyes are dark, slowly return to the east (angle 10) ready for sunrise.
- **Light it up:** connect an LED to the solar panel. Does it glow brighter when tracking?
- **Save power:** check only once a second instead of every 30 ms. Does it still work?
- **Data scientist:** print left, right and angle to the Serial plotter and watch the graph.

Think like an engineer

Our tracker turns left and right. Real big trackers often also tilt **up and down**. How many servos and LDRs would that need?

Boss level

Answer: 2 servos and 4 LDRs in a + shape!

Show what you know

What did we learn?

- [x] Straight-on sunlight gives a panel the most energy
- [x] Two LDRs plus a shadow wall tell us where the light is
- [x] The servo turns a little towards the brighter side
- [x] Feedback: sense, compare, move, repeat
- [x] A dead zone stops the robot from twitching

Exit ticket: why do we need a wall between the two LDRs?

At home

At 9 am, 12 pm and 4 pm, look where the sun is. Draw its path across the sky.



Next class: a Home Burglar Alarm with a Secret Code!